

COMANCHE, TX, USA

WMO: 747355

Lat: 31.917N

Lon: 98.600W

Elev: 423

StdP: 96.34

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53997

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.0	(c) -2.4	(d) -13.0	(e) 1.3	(f) 2.7	(g) -11.3	(h) 1.5	(i) 5.0	(j) 12.0	(k) 11.2	(l) 11.1	(m) 12.0	(n) 4.1	(o) 0	(p) 0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.0	(c) 37.8	(d) 22.1	(e) 37.0	(f) 22.1	(g) 35.9	(h) 22.3	(i) 25.1	(j) 31.7	(k) 24.6	(l) 31.0	(m) 24.1	(n) 30.4	(o) 4.0	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	19.6	26.4	23.0	18.7	26.0	22.7	18.3	25.9	78.8	32.1	76.7	30.6	75.0	30.1	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.1	8.2	-8.6	39.8	1.8	1.2	-9.9	40.7	-10.9	41.5	-11.9	42.2	-13.2	43.1
			-9.4	25.9	2.0	0.9	-10.8	26.5	-12.0	27.0	-13.1	27.5	-14.5	28.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	8.8	10.9	15.3	19.0	22.7	27.2	28.8	28.8	25.5	19.5	14.2	9.5
	DBStd	8.43	5.77	6.34	5.26	4.43	3.65	2.34	2.24	2.52	3.12	4.96	5.19	5.56
	HDD10.0	274	91	61	16	3	0	0	0	0	0	4	21	79
	HDD18.3	1151	297	217	121	43	9	0	0	0	1	46	140	277
	CDD10.0	3641	55	84	181	274	393	516	582	583	465	298	146	64
	CDD18.3	1474	3	8	26	64	144	266	323	324	216	82	16	3
	CDH23.3	16357	30	78	234	597	1312	3046	4001	3963	2182	756	126	31
CDH26.7	8108	4	17	57	203	518	1541	2210	2207	1053	267	26	4	
Wind	WSAvg	3.9	4.1	4.4	4.6	4.6	4.1	3.9	3.4	3.1	3.0	3.6	3.8	3.9
Precipitation	PrecAvg	850	40	73	58	73	95	99	59	72	71	95	56	47
	PrecMax	1223	93	190	113	258	283	190	169	297	132	181	180	212
	PrecMin	534	5	2	1	1	23	13	0	0	6	20	0	3
	PrecStd	177	24	49	34	60	56	53	47	78	42	57	53	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.9	28.9	30.9	34.7	35.2	38.1	39.5	39.2	37.5	34.3	29.7	26.8
		MCWB	13.7	16.3	17.0	18.3	22.2	22.1	21.8	22.8	21.3	21.1	19.1	15.3
	2%	DB	23.0	25.9	28.0	31.3	33.0	36.6	37.7	37.8	35.4	31.7	27.1	23.0
		MCWB	13.6	15.2	16.8	18.3	21.8	22.0	21.9	22.5	22.2	21.2	17.2	13.8
	5%	DB	20.9	22.9	26.2	28.8	31.8	35.0	36.6	37.1	33.9	29.8	24.1	21.0
		MCWB	12.9	14.3	16.3	18.1	21.1	22.4	22.4	22.4	22.0	20.3	16.2	13.9
	10%	DB	17.8	20.9	23.6	27.1	29.8	33.4	35.2	35.8	32.4	27.7	22.3	18.0
		MCWB	11.5	13.7	15.6	18.1	20.9	22.5	22.6	22.5	21.9	18.7	15.7	12.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	19.6	20.4	22.6	24.8	25.9	25.7	25.5	25.0	23.8	21.4	19.2
		MCDB	21.3	23.2	25.8	27.3	30.1	32.1	32.7	32.0	30.8	28.3	25.5	21.8
	2%	WB	16.3	18.4	19.5	21.5	23.9	24.9	25.0	24.9	24.2	22.9	19.9	17.5
		MCDB	19.7	22.1	24.0	26.5	29.0	31.0	31.7	32.0	29.9	27.5	22.9	20.2
	5%	WB	14.2	16.9	18.6	20.6	23.1	24.2	24.4	24.4	23.8	22.1	18.7	15.6
		MCDB	18.6	20.4	22.9	25.6	28.2	29.9	31.1	31.4	29.3	26.5	21.5	18.4
	10%	WB	12.1	15.2	17.6	19.8	22.4	23.7	23.9	23.9	23.2	21.2	17.6	13.7
		MCDB	17.5	19.1	21.4	24.2	27.3	29.6	30.5	30.8	28.4	25.6	20.7	17.3
Mean Daily Temperature Range	5% DB	MDBR	11.8	12.2	12.4	12.5	11.3	11.6	12.0	12.3	11.7	12.4	12.2	11.7
		MCDBR	14.9	15.1	14.8	15.1	13.9	13.4	13.4	13.7	13.7	13.5	13.7	14.9
		MCWBR	8.0	8.2	6.7	6.8	5.1	3.6	3.5	3.4	3.9	5.5	7.0	8.2
	5% WB	MCDBR	13.1	13.2	11.1	11.4	11.2	10.8	11.5	11.8	11.1	11.5	11.0	12.0
		MCWBR	9.0	8.8	6.4	6.7	5.0	4.0	3.7	3.6	3.8	6.0	7.9	8.7
Clear-Sky Solar Irradiance	taub	0.301	0.310	0.336	0.375	0.412	0.413	0.423	0.415	0.387	0.352	0.322	0.305	
	taud	2.521	2.505	2.446	2.360	2.318	2.352	2.370	2.375	2.396	2.482	2.496	2.517	
	Ebn at Noon	929	953	945	915	879	873	862	867	880	893	895	904	
	Edh at Noon	85	95	109	124	130	125	122	120	113	97	87	81	
All-Sky Solar Radiation	RadAvg	3.10	3.73	4.81	5.81	6.10	6.86	6.87	6.28	5.28	4.20	3.26	2.75	
	RadStd	0.33	0.55	0.39	0.50	0.60	0.56	0.49	0.43	0.44	0.63	0.42	0.36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A

Nomenclature: See separate page